Cooling Load Calculation Formula

Step 1: Base Cooling Load

$$\text{Base Load (BTU/hr)} = \text{Room Area (sqm)} \times 337$$

*337 BTU/hr per sqm is used for tropical climate conditions.

Step 2: Adjustment Factors

a. Ceiling Height Adjustment

$$(\text{Ceiling Height} - 2.5) \times 100$$

*Standard ceiling height = 2.5 meters

b. Occupant Heat Load

$$\text{Number of Occupants} \times 600$$

* Each person contributes approximately 600 BTU/hr.

c. Sunlight Exposure Adjustment

$$\text{Sunlight Level} \times 500$$

Sunlight Exposure Value

Low=1

Medium=2

High=3

Step 3: Total Cooling Load

$$\text{Total Cooling Load (BTU/hr)} = (\text{Area} \times 337) + ((\text{Ceiling Height} - 2.5) \times 100) + (\text{Occupants} \times 600) + (\text{Sunlight Level} \times 500)$$

Step 4: Convert BTU/hr to Horsepower (HP)

$$\text{Required HP} = \frac{\text{Total BTU/hr}}{9000}$$

*1 HP = 9,000 BTU/hr

Step 5: Final HP Recommendation

$$\text{Final HP} = \text{Round UP to nearest 0.5 HP}$$

Step 6: Determine Unit Type (Inverter / Non-Inverter)

Unit Type Recommendation

Inverter = Usage Hours >= 6

Non Inverter = Usage Hours < 6

Sample Cooling Load Computation

Given:

Room Area: 20 sqm

Ceiling Height: 3.0 meters

Number of Occupants: 3 persons

Sunlight Exposure: Medium (Level 2)

Estimated Daily Usage = 7 hours/day (Frequent Usage)

Step 1: Base Cooling Load

20×337=6,740 BTU/hr

Step 2: Ceiling Height Adjustment

(3.0−2.5)×100=50 BTU/hr

Step 3: Occupant Heat Load

3×600=1,800 BTU/hr

Step 4: Sunlight Exposure Adjustment

2×500=1,000 BTU/hr

Step 5: Total Cooling Load

6,740+50+1,800+1,000=9,590 BTU/hr

Step 6: Convert BTU/hr to HP

$$\frac{9{,}590}{9000}=1.07 \text{ HP}$$

Step 7: Final HP Recommendations

Round UP to nearest 0.5 HP=1.5 HP

HP Needed = 1.5

Step 8: Determine Unit Type (Based on Usage)

Estimated Daily Usage=7 hours/day (Frequent)

Unit Type Recommendation=INVERTER